

# Personalized AI assistants could either promote or hinder the emergence of collective intelligence

Jaeseung Kim<sup>1</sup> Sang Woo Park<sup>1,2,\*</sup>

**1** School of Biological Sciences, Seoul National University, Seoul, Korea

**2** Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

\*Corresponding author: sangwoopark@snu.ac.kr

## Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

## Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI services on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI services on human CI is hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of indentifying a robust incentive structure that can promote CI in the presence of AI, which can retain the benefits of AIs while preventing potential risk such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. We allow each individual to independently consult with personal AIs and revise their predictions accordingly. Across different assumptions about the AI model’s ability and underlying incentive structures, we investigate the robustness of human CIs.

## Main

## Model

We begin by extending previously established models of CI in which a group of players participates in a prediction task, with each player making an individual prediction based on partial observations of the underlying causal factors and the group forming a collective prediction by aggregating these individual predictions [30, 12, 13]. The prediction target ( $Y$ ) is modeled as a linear combination of  $M$  causal factors ( $x_i$ ), with fixed coefficients ( $\alpha_i$ ):  $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$ , where each factor is drawn from independent normal distribution. Among  $M$  different factors, an individual ( $A$ ) makes prediction by the multiplication of single factor of interest ( $x_{e_A}$ ) and their own belief ( $c_A$ ):  $Y_A = c_A x_{e_A}$ . Here, we assume each individual can independently consult AI and revise their prediction based on their reliance on AI ( $\beta_A$ ):  $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ , and the revised prediction is aggregated to form collective prediction.

To model AI, we consider a scenario where the group of players is given a novel prediction task, whereas the AI has an ability to perform prediction based on causal factors because it already been trained on similar tasks. Every time a player consults AI, it will make predictions ( $Y_A^{AI}$ ) based on its internal parameters, which can include factor-specific bias ( $b_i$ ) and stochastic error ( $\epsilon$ ). We then consider two extreme types of AI response, which differ in the amount of information AI uses when making predictions. At one extreme, the AI operates with complete knowledge of all causal factors, using them regardless of the user’s interest:  $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$ . At the other extreme, AI can work like a chatbot, in which AI predicts the outcome based only on the user’s interest:  $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$ . Hereafter, we refer to these two types as ‘Omniscient AI’ and ‘Chatbot AI’, respectively.

Each generation consists of multiple round of predictions, and each player is rewarded by a payoff according to the incentive structure. For simplicity, we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff rather than computing the payoff for each round. We consider two incentive structures, “feedback” and “niche-expert” [13], which have been shown to promote near-perfect collective accuracy in the absence of AI (except averaging aggregation with niche-expert structure). The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy. In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest.

Through time, the group evolves by imitating the peers who receive higher payoffs. Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (factor of interest, belief, reliance on AI) of the other with a probability determined by the difference between their expected payoffs. Individual predictions are aggregated into a collective prediction ( $\hat{Y}$ ) by simple averaging when

103 using Omniscient AI and by clustering when using Chatbot AI. Collective accuracy  
 104 is defined as the reduction in prediction error achieved by the collective prediction  
 105 relative to a baseline prediction ( $\alpha_0$ ). Hereafter, we refer to the near-perfect collective  
 106 accuracy as accurate CI.

107 This model now allows us to explore the impact of AI on CI depending on the  
 108 mode of AI and the incentive structure. Here, we begin by comparing the emergence  
 109 of CI with and without AI and subsequently investigate a better incentive structure  
 110 that can promote CI in the presence of AI.

## 111 **AI can either promote or prevent the emergence of CI**

112 To understand the impact of AI, we first compare collective dynamics with and  
 113 without AI. We find that, except when Omniscient AI is used under the feedback  
 114 structure, a group with AI fails to achieve accurate CI (Fig. 2). However, the effect  
 115 of AI differs depending on the mode of AI.

116 When Omniscient AI assists players, the group becomes equally or more accurate  
 117 than the group without AI. Under the feedback structure, a group achieves accurate  
 118 CI regardless of the assistance of AI (Fig. 2A). However, under the niche-expert  
 119 structure, the group without AI is no better than the baseline prediction. Although  
 120 Omniscient AI cannot help players achieve accurate CI, the group’s prediction with  
 121 AI is more accurate than both that of the group without AI and the AI itself (Fig.  
 122 2B). These results are robust across different levels of AI accuracy and can be shown  
 123 mathematically by assuming a sufficiently large group of players (Fig. S2).

124 When Chatbot AI assists players, the group becomes less accurate than the group  
 125 without AI. Although both incentive structures can achieve accurate CI without AI,  
 126 the assistance of Chatbot AI hinders group performance and prevents accurate CI  
 127 from emerging (Fig. 2C, D). This occurs through different mechanisms depending  
 128 on the incentive structure. Under the feedback structure, Chatbot AI leads to a loss  
 129 of interest diversity, which eventually makes it impossible for the group to account  
 130 for variation in some causal factors. This leads to an increase in collective variance  
 131 and stochastic decrease in collective accuracy (Fig. S1). The loss of interest diversity  
 132 may arise even without AI after sufficiently many generations in a finite population.  
 133 However, Chatbot AI makes the group more vulnerable to the diversity loss. Un-  
 134 der the niche-expert structure, the group becomes overly reliant on AI, causing the  
 135 collective prediction to inherit the AI’s bias (Fig. S1). This leads to an increase in  
 136 collective bias and a decrease in collective accuracy. The niche-expert structure may  
 137 also drive reliance on AI toward zero, depending on the relationship between the  
 138 true coefficient and AI bias (Fig. S2). In this case, the group completely excludes  
 139 AI assistance and builds CI on its own, which substantially delays the emergence of  
 140 accurate CI.

141 To investigate whether the group can retain its predictive ability when AI as-  
 142 sistance suddenly becomes unavailable at a given generation, we measure counter-  
 143 factual collective accuracy. This measure follows the same evolutionary trajectory

144 as the simulation with AI, but accuracy is computed using the players’ unrevised  
 145 predictions. We find that counterfactual collective accuracy is always lower than the  
 146 collective accuracy of groups, regardless of the AI mode or incentive structure.

147 In summary, AI assistance may promote CI when it is omniscient, whereas AI  
 148 operating with limited information may prevent the emergence of accurate CI. These  
 149 findings underscore the need for incentive structures that can mitigate the potential  
 150 risks associated with AI assistance.

## 151 **Directly incentivizing or penalizing AI use can additionally** 152 **harm CI**

153 Given the strong influence of AI on collective outcomes, a straightforward interven-  
 154 tion is to directly incentivize or penalize the use of AI. To investigate whether such  
 155 direct incentives or penalties help emergence of CI, we modify the two incentive  
 156 structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

157 where  $\lambda$  represent the strength of incentive ( $\lambda > 0$ ) or penalty ( $\lambda < 0$ ).

158 Directly incentivizing or penalizing AI use further reduce the collective accuracy  
 159 (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure  
 160 (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the population  
 161 to always ( $\beta = 1$ ) or never ( $\beta = 0$ ) rely on AI, respectively (Fig. 3B, D, F, H). In some  
 162 cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can  
 163 achieve higher collective accuracy than the original incentive structure. However, in  
 164 most cases, penalizing AI use imposes strong selection against AI reliance, pushing  
 165 the group toward lower stationary collective accuracy. Although the group with  
 166 Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn’t  
 167 result in stationary collective accuracy, incentivizing or penalizing AI use further  
 168 strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing  
 169 or penalizing the AI use may exacerbate rather than resolving the problem of AI.

## 170 **Balanced incentives mitigate AI-induced failures of CI**

171 Neither the original incentive structures nor simple interventions targeting AI use  
 172 are sufficient to promote CI. What incentive structure, then, would work in the  
 173 presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures  
 174 fail to promote accurate CI, we investigate whether a better structure can overcome  
 175 the risks introduced by AI. Here, we propose a “balanced” structure, defined as the  
 176 simple average of the feedback and niche expert incentives.

177 Core idea behind the balanced structure is to maintain interest diversity while  
 178 preventing the collective bias of the group. The feedback structure is prone to in-  
 179 terest diversity loss, whereas can strongly prevent collective bias by incentivizing  
 180 players whose predictions lie opposite to the collective prediction. By contrast, the

181 niche-expert structure can maintain interest diversity because it directly incentivizes  
182 minority interests. However, it cannot prevent collective bias because, within the  
183 same factor of interest, individuals are rewarded only according to their individual  
184 predictive performance, which homogenize the predictions.

185 By integrating both structure, the balanced structure can promote accurate CI by  
186 maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Com-  
187 pare to the feedback and niche-expert, players are both evenly distributed through  
188 factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfac-  
189 tual collective accuracy maintained high compare to both feedback and niche-expert.  
190 These patterns are driven by the emergent property of the balanced structure, in  
191 which group increases its reliance on AI as the AI becomes more accurate, demon-  
192 strating appropriately calibrated AI use.

## 193 Discussion

194 In this paper, we study how AI and incentive structures affect the emergence of CI.  
195 Our model suggests that the effect of AI depends on the type of AI used by the  
196 group. AI that can incorporate all relevant causal factors may improve group perfor-  
197 mance, whereas AI that relies on users to gather information about the causal factors  
198 may impair it. By examining different variants of incentive structures, we further  
199 show that direct interventions targeting AI use generally harm group performance.  
200 Instead, we suggest balanced structure which can promote CI while also generating  
201 appropriately calibrated AI use as an emergent property.

202 Although Omniscient AI can produce group performance that is equal to or bet-  
203 ter than that achieved without AI, such AI may be difficult to realize in practice  
204 because it needs to be able to observe causal factors independently. Achieving this  
205 capability would require hardwares, such as sensors, as well as autonomous data ac-  
206 quisition capabilities. For this reason, most AI services currently available are better  
207 represented by the Chatbot AI in our model, which highlights the potential risks of  
208 AI use for CI.

209 In our model, individuals revise their predictions based on the AI’s assistance.  
210 It is therefore unsurprising that AI assistance reduces the diversity of individual  
211 predictions, a pattern that has also been observed empirically. However, a reduction  
212 in prediction diversity does not necessarily lead to the collapse of CI. The principal  
213 risk arises when the incentive structure fails to guide the group toward an appropriate  
214 set of strategies.

215 Because AI already generates substantial benefits as well as costs, organizations  
216 increasingly seek to improve collective performance by regulating AI use. Such in-  
217 terventions may take direct forms, including AI-use policies or token-based leader-  
218 boards, or may operate indirectly through other incentives, such as productivity  
219 rewards or the social stigma associated with AI use. However, our model suggests  
220 that directly incentivizing or penalizing AI use is unlikely to induce appropriately

calibrated reliance on AI or to promote CI. Incentivizing AI use may lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use may lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our results instead suggest that appropriate AI use and accurate CI can be promoted by focusing on the classic properties of groups that have been emphasized for decades—maintaining diversity and preventing collective bias—rather than on AI use itself.

Limited level of counterfactual collective accuracy observed across AI types and incentive structures suggests that the presence of AI can degrade human group’s own performance. This may reduce collective performance when AI assistance becomes unavailable because of resource constraints, system failure, or model collapse, forcing the group to rebuild CI through further adaptation. The higher counterfactual collective accuracy achieved under the balanced structure indicates that an appropriate incentive structure can mitigate the risk of CI breakdown following the loss of AI assistance.

Our model and analysis have several limitations. First, the model considers only social learning among humans and treats the influence of AI as unidirectional. In reality, humans can learn from AI, while AI systems can also be further trained on human-generated data. Such bidirectional adaptation may further reduce group performance by increasing the similarity between human and AI predictions and thereby weakening their complementarity. Empirical studies of human–AI collaboration likewise suggest that stronger correlations between human and AI predictions can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random noise. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

## Methods

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target  $Y$  is modeled as a linear combination of  $M$  independent factors  $(x_1, x_2, \dots, x_m)$  and  $M + 1$  coefficients  $(\alpha_0, \alpha_1, \dots, \alpha_m)$ , where  $\alpha_0$  denotes the global offset (Fig. 1A). The  $M$

261 factors are randomly drawn from normal distributions with mean  $X_i$  and standard  
 262 deviation  $\sigma_i$  in each generation.

263 Among a population of  $N$  individuals, each individual can observe a single factor  
 264 and make a prediction based on the observed value and their own belief. In particular,  
 265 individual  $A$ 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (2)$$

266 where  $c_A$  represents individual  $A$ 's personal belief,  $e_A$  represents the factor that  $A$   
 267 observes (interest), and  $x_{e_A}$  represents the realized value of  $A$ 's interest. Individuals  
 268 can also choose not to observe any factor, which represents stubborn individuals  
 269 who make predictions only based on their own belief. For notational convenience,  
 270 we assume that stubborn individuals have interest 0, and let  $x_0$  always be 1.

271 Next, we model AI as a prediction system that has learned the true coefficients  
 272 with bias and error (Fig. 1B). For each factor  $i \in [0, M]$ , AI knows the exact  
 273 coefficient  $\alpha_i$  and has a fixed bias  $b_i$ , resulting in a prediction of  $\alpha_i x_i + b_i$ . In terms  
 274 of human-AI interaction, we model two contrasting modes of AI: the "AI knows all"  
 275 model and the "AI answers question" model. In the AI knows all model, AI acts  
 276 as an omniscient system and gives an answer regardless of the individual's interest,  
 277 accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (3)$$

278 In contrast, the AI answers question model acts as a chatbot and gives an answer  
 279 only based on the individual's interest. Thus, given the individual's interest,  $e_A$ , AI  
 280 predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (4)$$

281 In both models,  $\epsilon \sim N(0, \sigma_\epsilon^2)$  represents random error arising from the stochastic  
 282 generative process of the AI model.

283 AI works as a personalized assistant for each individual, such that each individual  
 284 receives assistance by incorporating the AI's prediction. However, even with the same  
 285 opportunity to receive AI assistance, reliance on AI can vary substantially across  
 286 individuals. To model this, we assume that the revised prediction is a weighted sum  
 287 of the individual's own prediction and the AI's prediction by introducing AI belief  
 288  $\beta_A$  as a weighting coefficient. The revised prediction of individual  $A$  can be written  
 289 as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (5)$$

290 The revised predictions of individuals are aggregated by two different mechanisms,  
 291 depending on the mode of AI. In the AI knows all model, even if an individual can  
 292 observe only a single factor, their revised prediction contains information about all

293 factors through interaction with AI. Therefore, we use simple averaging of revised  
 294 predictions as the collective prediction. In contrast, in the AI answers question  
 295 model, each individual gives a prediction based only on a single factor even after  
 296 revising their prediction with AI assistance. Therefore, we use clustering aggregation  
 297 to obtain the collective prediction, which is performed by clustering individuals based  
 298 on interest and summing the average values of each cluster.

299 For each generation, each individual is rewarded with a payoff based on the given  
 300 incentive structure (Fig. 1C). We adopt two incentive structures, “feedback” and  
 301 “niche expert,” which were proposed by Wang et al. [13] and promote collective  
 302 intelligence when prediction is performed without AI. The feedback incentive gives  
 303 higher payoff to individuals who can push the collective prediction toward the true  
 304 value, regardless of their individual performance. Given individual  $A$ ’s revised pre-  
 305 diction  $Y_A^{revised}$  and the collective prediction  $\hat{Y}^{revised}$ , it can be expressed as:

$$\pi_A = Y_A^{revised}(Y - \hat{Y}^{revised}) \quad (6)$$

306 The niche expert incentive gives higher payoff to individuals who have both better  
 307 individual performance and a minority interest. Specifically, the niche expert payoff  
 308 is expressed as the product of the proportion of individuals who observe the same  
 309 factor as individual  $A$ ,  $\rho_{e_A}$ , and individual accuracy. Given individual  $A$ ’s revised  
 310 prediction  $Y_A^{revised}$ , it can be expressed as:

$$\pi_A = -\rho_{e_A}(Y_A^{revised} - Y)^2 \quad (7)$$

311 These two incentive structures were shown to achieve nearly perfect collective in-  
 312 telligence through social learning (except for the niche expert incentive with simple  
 313 averaging aggregation) by promoting individual expertise while maintaining collec-  
 314 tive diversity. Therefore, we use these incentive structures as a baseline to under-  
 315 stand how AI affects the emergence of collective intelligence. Moreover, we modify  
 316 the two baseline incentives in a later section to understand the consequences of direct  
 317 incentives and penalties for AI usage.

318 Throughout the simulation, individuals engage in social learning by imitating  
 319 other individuals who receive higher payoffs. We assume that prediction events  
 320 occur more frequently than social learning events within the population, such that  
 321 individual payoff can be computed as an expected payoff. The imitation process  
 322 is performed for at most one pair in each generation. By sampling two random  
 323 individuals,  $A$  and  $B$ , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (8)$$

324 where  $s$  denotes the strength of social learning. When imitation occurs,  $A$  imitates  
 325 all three strategies of  $B$ : interest, belief, and AI belief.

326 To evaluate and analyze the dynamics of collective prediction, we define six sum-  
 327 mary metrics: collective accuracy, collective bias, collective variance, human accu-  
 328 racy, interest diversity, and median AI belief. These metrics capture the accuracy

of the revised collective prediction, its bias–variance decomposition, the predictive performance without AI, the diversity of individual interests, and median AI belief in the population (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

| Metric                             | Definition                                                                              | Interpretation                                                                                    |
|------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Collective accuracy                | $1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$ | Accuracy of the revised collective prediction relative to the baseline prediction $\alpha_0$ .    |
| Collective bias                    | $\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$                               | Squared systematic deviation of the revised collective prediction from the true value.            |
| Collective variance                | $\text{Var}(Y - \hat{Y}^{\text{revised}})$                                              | Variability of the revised collective prediction error across prediction events.                  |
| Counterfactual collective accuracy | $1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$                  | Accuracy of the human-only collective prediction relative to the baseline prediction $\alpha_0$ . |
| Interest diversity                 | $\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$                                      | Effective number of interests represented in the population.                                      |
| Median reliance on AI              | $\text{median}_{A \in N}(\beta_A)$                                                      | Median value of reliance on AI $\beta_A$ in the population.                                       |

## Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

## References

- [1] Francis Galton. Vox populi, 1907.
- [2] Anita Williams Woolley, Christopher F Chabris, Alex Pentland, Nada Hashmi, and Thomas W Malone. Evidence for a collective intelligence factor in the performance of human groups. *science*, 330(6004):686–688, 2010.

- [3] Ilan Yaniv and Maxim Milyavsky. Using advice from multiple sources to revise and improve judgments. *Organizational Behavior and Human Decision Processes*, 103(1):104–120, 2007.
- [4] Cars Hommes, Joep Sonnemans, Jan Tuinstra, and Henk Van de Velden. Coordination of expectations in asset pricing experiments. *The Review of Financial Studies*, 18(3):955–980, 2005.
- [5] Irving Lorge, David Fox, Joel Davitz, and Marlin Brenner. A survey of studies contrasting the quality of group performance and individual performance, 1920–1957. *Psychological bulletin*, 55(6):337, 1958.
- [6] Tabea Dreyer, Amir Haluts, Amos Korman, Nir Gov, Ehud Fonio, and Ofer Feinerman. Comparing cooperative geometric puzzle solving in ants versus humans. *Proceedings of the National Academy of Sciences*, 122(1):e2414274121, 2025.
- [7] Andrew M Simons. Many wrongs: the advantage of group navigation. *Trends in ecology & evolution*, 19(9):453–455, 2004.
- [8] Andrew Berdahl, Colin J Torney, Christos C Ioannou, Jolyon J Faria, and Iain D Couzin. Emergent sensing of complex environments by mobile animal groups. *Science*, 339(6119):574–576, 2013.
- [9] Christoph Riedl, Young Ji Kim, Pranav Gupta, Thomas W Malone, and Anita Williams Woolley. Quantifying collective intelligence in human groups. *Proceedings of the National Academy of Sciences*, 118(21):e2005737118, 2021.
- [10] Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996.
- [11] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.
- [12] Richard P Mann and Dirk Helbing. Optimal incentives for collective intelligence. *Proceedings of the National Academy of Sciences*, 114(20):5077–5082, 2017.
- [13] Guocheng Wang, Qi Su, Long Wang, and Joshua B Plotkin. Individual incentives that promote collective intelligence. *Proceedings of the National Academy of Sciences*, 122(51):e2516535122, 2025.
- [14] Justin Werfel. Fluid thinking about collective intelligence. *Nature Machine Intelligence*, 8(4):506–516, 2026.
- [15] Hao Cui and Taha Yasseri. AI-enhanced collective intelligence. *Patterns*, 5(11):101074, 2024.
- [16] Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of AI research. *Nature*, 651(8107):914–919, 2026.

- [17] Anil R Doshi and Oliver P Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.
- [18] Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarlaschi, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, et al. Artificial intelligence index report 2026. *arXiv preprint arXiv:2606.15708*, 2026.
- [19] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8(12):2293–2303, 2024.
- [20] Jason W Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A Bakker, Joshua A Becker, Aleks Berditschevskaia, Julian Berger, Levin Brinkmann, et al. How large language models can reshape collective intelligence. *Nature human behaviour*, 8(9):1643–1655, 2024.
- [21] Lennart Meincke, Gideon Nave, and Christian Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature human behaviour*, 9(6):1107–1109, 2025.
- [22] Andreas Fügener, Jörn Grahl, Alok Gupta, and Wolfgang Ketter. Will humans-in-the-loop become borgs? merits and pitfalls of working with AI. *MIS quarterly*, 45(3):1527–1556, 2021.
- [23] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.
- [24] Lucy M Aplin, Damien R Farine, Richard P Mann, and Ben C Sheldon. Individual-level personality influences social foraging and collective behaviour in wild birds. *Proceedings of the Royal Society B: Biological Sciences*, 281(1789):20141016, 2014.
- [25] Lu Hong and Scott E Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.
- [26] Isabelle Bousquette. Why some companies say AI ‘tokenmaxxing’ is key to survival. <https://www.wsj.com/cio-journal/why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128>, April 2026. The Wall Street Journal. Accessed: 2026-07-07.
- [27] Tim Keary. Is the cult of ‘tokenmaxxing’ just another fad or the new normal? <https://www.forbes.com/sites/timkeary/2026/04/13/is-the-cult-of-tokenmaxxing-just-another-fad-or-the-new-normal/>, April 2026. Forbes. Accessed: 2026-07-07.

- 412 [28] Dalmeet Singh Chawla. Researchers who use hallucinated references to face  
413 arXiv ban. *Nature*, 653(8116):988–989, 2026.
- 414 [29] H Holden Thorp. Chatgpt is fun, but not an author. *Science*, 379(6630):313–313,  
415 2023.
- 416 [30] Lu Hong, Scott E Page, and Maria Riolo. Incentives, information, and emergent  
417 collective accuracy. *Managerial and Decision Economics*, 33(5-6):323–334, 2012.

## 418 Acknowledgements

419 S.W.P. was supported by the New Faculty Startup Fund from Seoul National Univer-  
420 sity, and the Global-LAMP Program of the National Research Foundation of Korea  
421 (NRF) grant funded by the Ministry of Education (No. RS-2023-00301976).

## 422 Contributions

423 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,  
424 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &  
425 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-  
426 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

## 427 Competing Interests

428 The authors declare no competing interests.

## 429 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ( $n=52$ ). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ( $n=52$ ). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ( $n=8000$  posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ( $n=8000$  posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected  $i(0)$  and susceptible  $S(0)$  at the beginning of season on center of gravity. Points represent the estimated values for  $i(0)$  and  $s(0)$  between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the  $i(0)$  value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of  $i(0)$ . (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from  $n=8000$  posterior samples. Shaded regions represent the 95% credible interval in our estimates ( $n=8000$  posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from  $n=8000$  posterior samples. Error bars represent the 95% credible interval in our estimates ( $n=8000$  posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.